

求积分

$$\int_0^1 \ln x \cdot \ln(1-x) dx$$

首先引入两个引理：

引理一：

$$\begin{aligned} & \int_0^1 x^n \ln x dx \\ &= \frac{x^{n+1}}{n+1} \ln x - \frac{1}{n+1} \int_0^1 x^n dx \\ &= -\frac{1}{(n+1)^2} \end{aligned}$$

引理二：

$$\zeta(2) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

原积分展开得到

$$\begin{aligned} & -\sum_{n=1}^{\infty} \frac{1}{n} \int_0^1 x^n \ln x dx \\ &= \sum_{n=1}^{\infty} \frac{1}{n(n+1)^2} \\ &= \sum_{n=1}^{\infty} \frac{1}{n} - \frac{1}{n+1} - \frac{1}{(n+1)^2} \\ &= 1 + 1 - \zeta(2) = 2 - \frac{\pi^2}{6} \end{aligned}$$